LANS-RISK: Queue-Aware Supervision of Learned Driving Policies under Actuation Delay

Rizwan Ali and Jie Tian
School of Automotive and Traffic Engineering
Nanjing Forestry University, Nanjing, China

***Abstract*—Commands selected from a current vehicle state can become inappropriate while waiting for delayed execution. This paper investigates a queue-aware, trajectory-retaining supervisor for frozen learned driving policies. LANS-RISK predicts the state reached after pending commands, incorporates a proximal policy optimization proposal into sampled future plans, and ranks plans by predicted collision-free duration, corridor retention, and tracking cost. We report a focused, 22,000-episode subset of a separately frozen confirmation study, covering four synthetic traffic scenarios, five command delays, a matched nominal model predictive controller, PPO-only control, and two component ablations. LANS-RISK and nominal MPC attain identical observed contiguous passing delay thresholds: 1,000 ms on Highway and Intersection, 700 ms on Merge, and 500 ms on Roundabout. Removing queue prediction reduces corridor completion in 45 of 60 paired actor/scenario/delay contrasts, all significant after the retained study-wide Holm correction. Removing the actor does not change corridor-completion rates on the evaluated grid. Tracking differences against MPC remain scenario-dependent. These findings support queue compensation within this implementation, but do not establish a general advantage of learned proposals, universal superiority over MPC, or a formal safety guarantee.**

***Index Terms*—autonomous driving, actuation delay, predictive supervision, learned control, model predictive control, reproducible evaluation**

I. Introduction

Autonomous-driving control decisions are often evaluated as if the selected action were applied immediately. With a physical command queue, the vehicle instead continues to execute previously issued commands. A corrective action based only on the measured state can therefore address a state that no longer exists when the command reaches the plant. Nearby traffic also evolves during this interval. Delay compensation must consequently be evaluated together with collision outcomes, task completion, and tracking, rather than inferred from a controller's decision time alone.

Delay-aware control and supervision of learning-enabled controllers are established research directions. Kalaria *et al.* combine delay handling with robust tube MPC and discuss a protective role for a learning-enabled primary controller [1]. Predictive safety filtering also provides a principled framework for modifying a learned action under specified model and feasibility assumptions [2]. These precedents rule out a broad claim that predicting through delay or guarding a learned policy is itself new.

The author-supplied LANS path-tracking manuscript [4] studies analytical feedback with optional PPO steering residuals, packet-age handling, and implementation-specific supervision. The present study addresses a different question: how does a trajectory-level supervisor perform in interacting synthetic traffic when all commands, including corrective commands, pass through the same actuation FIFO? The current implementation combines existing model-based and learning-based elements. Its contribution is a specific design and a controlled examination of its benefits and limitations, not a new PPO optimization algorithm or a certified safety-filter theory.

We study LANS-RISK, a supervisor that rolls out the pending queue, retains a shifted previous trajectory, and includes a frozen PPO proposal as a prior in sampled trajectory search. Candidate selection first favors a longer predicted collision-free prefix, then corridor retention, and finally a weighted performance objective. This ordering is intended to distinguish collision-related recovery from a simple tracking-error penalty, while remaining an approximate forecast.

The conference contribution is deliberately focused: (i) an executable queue-aware trajectory-selection design with explicit information and actuation assumptions; (ii) a paired comparison with nominal MPC and PPO-only control; and (iii) matched removals of queue prediction and the actor proposal. We retain all four scenarios and all five command delays for the selected methods, including failures and ties. Broader controller families, plant/traffic perturbations, stochastic observation-delay evaluation, and detailed timing analyses are outside this paper's scope.

II. Related Work and Positioning

A. Delay-Aware Model-Based Control

The reference framework of Kalaria *et al.* addresses computation, command-processing, and actuator-related delays using delay estimation and robust control [1]. Our experiment uses a known, fixed-length command FIFO. It does not reproduce that framework's adaptive delay estimator, robust tube construction, racing experiments, or theoretical guarantees. This distinction matters: success under a known queue is not evidence of robustness to arbitrary unknown latency or uncertainty.

B. Learning and Predictive Supervision

PPO supplies a learned action proposal through the standard policy-optimization procedure [3]; no policy retraining occurs during the reported confirmation stage. Predictive safety filters

modify learning-controller actions using model-based constraints and, in certified formulations, appropriate uncertainty and backup-feasibility machinery [2]. LANS-RISK uses finite candidate sampling without a terminal invariant set or a verified robust backup. A collision-free sampled rollout is therefore a model prediction, not a proof of future safety.

Relative to the earlier path-tracking study [4], this controller evaluates steering/acceleration trajectories and surrounding-vehicle clearance. The benchmarks and RMSE definitions differ, precluding direct numerical comparison.

III. Problem Formulation

A. Shared Plant and Delayed Execution

Let x_t denote the ego state, z_t the observed surrounding-vehicle state, and $u_t \in [-1,1]^2$ a normalized steering/acceleration command selected at time t. For a queue of d control steps, the applied action is a previously selected command; schematically,

$$x_{t+1} = F(x_t, z_t, u_{t-d}), \qquad \Delta t = 0.1 \text{ s}. \tag{1}$$

The actual simulator also updates the interacting background vehicles. For $d = 0$, the new action is applied immediately. For $d > 0$, all methods share the same initially zero-filled FIFO. Neither supervision nor fallback bypasses this physical queue.

The benchmark uses a kinematic bicycle plant and synthetic disc-vehicle collision geometry. It is not a validated road-traffic simulator. Episodes last at most 160 steps, or 16 s, and may end earlier. All methods share scenario goals, action limits, observation access, and the physical delay. The collision diameter is 4.846 m; the reference-corridor half-width used for evaluation is 1.75 m. Initial scenes satisfy the benchmark's rejection-sampling predicates. These predicates exclude specified initial overlaps and close-following configurations; they do not prove that every resulting traffic task is solvable.

The evaluated scenarios are Highway, Merge, Intersection, and Roundabout. Background traffic follows the simulator's interaction rules. Predictive controllers infer NPC acceleration from observed velocity differences; they are not supplied hidden NPC desired speeds. Forecast mismatch is therefore present even though the simulation exposes structured state information.

B. Arrival-State Prediction

Write the pending commands in execution order as $q_t = (q_t^0, \dots, q_t^{d-1})$. The supervisor constructs an approximate arrival state by sequential propagation:

$$(\hat{x}_t^0, \hat{z}_t^0) = (x_t, z_t), \tag{2}$$

$$(\hat{x}_t^{j+1}, \hat{z}_t^{j+1}) = \hat{F}(\hat{x}_t^j, \hat{z}_t^j, q_t^j), \quad j < d. \tag{3}$$

The same predictor propagates both ego and surrounding-vehicle states. NPC acceleration estimates are clipped to the implemented bounds and damped during prediction. A diagnostic records predicted clearance or corridor violations along the pending prefix. Such a violation concerns commands already queued: a newly selected action cannot retroactively change that prefix.

IV. Queue-Aware Trajectory Supervision

A. Candidate Construction and Retention

From the predicted arrival state, the controller searches trajectories of $H = 30$ steps. Steering is parameterized relative to geometric nominal steering, while acceleration is normalized to the common plant action scale. The frozen PPO action supplies a first-action proposal and a candidate prior; it does not set a higher reference-speed ceiling than MPC.

Each of three sampling rounds evaluates 192 candidates. Random perturbations are correlated in five-step blocks around a per-step mean trajectory. Explicit candidates include the current mean, cruise, braking, the actor prior, the shifted retained plan, and the best candidate found so far. The 24 selected elite plans update the mean and dispersion. This is a cross-entropy-style sampled search, not an exact solution of the constrained optimal-control problem.

After selection, the previous plan is shifted by one step and extended with a geometric-steering/full-braking tail. Retention preserves a candidate for the next decision, but it does not establish recursive feasibility: the plant and traffic can differ from the predictions, and the appended tail has no invariant-set certificate.

B. Risk and Tracking Ranking

Let $c_{k,h}$ be predicted disc clearance and $e_{k,h}$ lateral reference error for candidate k at predicted step h. Define the first predicted collision-margin and corridor violations as

$$h_c(k) = \min\big(\{h : c_{k,h} < b\} \cup \{H\}\big), \tag{4}$$

$$h_e(k) = \min\big(\{h : |e_{k,h}| > w\} \cup \{H\}\big), \tag{5}$$

where $b = 0.25$ m is the forecast buffer and $w = 1.75$ m. The secondary rank is $h_e(k)$ unless $h_c(k) = 0$, in which case first-step clearance is used. Candidates are ordered lexicographically by larger h_c, larger secondary rank, and smaller performance cost. When an immediate forecast collision is unavoidable among the samples, comparing clearance avoids treating a tracking score alone as an adequate recovery criterion.

The performance cost includes lateral and heading error, speed-reference error, control effort, a soft clearance penalty, penalties for predicted clearance/corridor violations, and a first-action distance from the PPO proposal. In compact notation,

$$J_k = \sum_{h=0}^{H-1} \ell(\hat{x}_{k,h}, \hat{z}_{k,h}, u_{k,h}) + \lambda_\pi \|u_{k,0} - u_t^{\mathrm{PPO}}\|^2. \tag{6}$$

The implemented actor weight is $\lambda_\pi = 0.5$ when the actor is enabled. The exact weights and clipping operations remain fixed in the supplied source. The supervisor applies the selected first command and logs whether the chosen forecast violates either criterion within the horizon. Logging a violation is not a guarantee that the controller can avoid it.

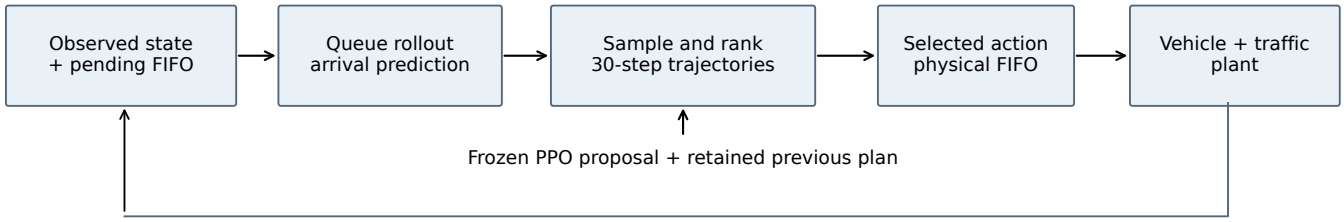


Fig. 1. Implemented information flow. Pending commands are rolled out before future trajectories are ranked. A frozen PPO proposal and the shifted previous plan enter candidate construction. Every selected action reaches the plant through the shared physical FIFO. The schematic does not imply certified safety or instantaneous control authority.

C. Matched Removals

The *no-prediction* variant disables internal pending-queue propagation while preserving the actual physical FIFO, actor, search budget, and selection mechanism. It evaluates the contribution of queue prediction within this design. The *no-actor* variant replaces the learned proposal with an analytical prior and removes the actor-distance penalty, retaining prediction, sampling, ranking, and trajectory retention. Its contrast measures the actor's combined prior/penalty contribution.

PPO-only control removes the entire supervisor. Consequently, LANS-RISK versus PPO-only is a comparison of a supervision bundle, not a clean causal test of one safety mechanism. None of these ablations establishes a general conclusion about all learned controllers or all prediction algorithms.

V. Experimental Protocol

A. Frozen Confirmation and Conference Scope

The reported subset contains five controller variants: LANS-RISK, nominal MPC, PPO-only, no queue prediction, and no actor. Each is evaluated on all four scenarios and delays of 0, 200, 500, 700, and 1,000 ms. Learned variants use frozen actor seeds 42, 43, and 44; actor-free methods are run once for each scenario/delay. Each condition uses the same 100 traffic-case seeds, 141000–141099. This yields 220 conditions and 22,000 episode records. The repeated actor evaluations reuse traffic cases and are not 300 independent cases per condition.

These records are drawn from a completed, source-frozen 440-condition/44,000-record confirmation stage. Development used a separate seed range. No retraining or tuning was performed on the reported confirmation cases. The conference subset was chosen for its methodological scope after the full evaluation; it is not a separately preregistered experiment. All selected methods retain every scenario and delay. The expanded robustness, sensor-delay, runtime, and broader baseline studies are not reported here. In particular, this paper uses only the original main-confirmation protocol, not the separate dense-Merge robustness initialization amendment.

B. Nominal MPC Comparison

The nominal MPC shares the plant, structured state information, physical FIFO, action bounds, reference-speed ceiling, observed-acceleration predictor, 30-step horizon, and three rounds of 192 candidates. It uses a block-parameterized plan and its own cost ranking, warm start, and fallback. LANS-RISK differs in actor use, trajectory retention/representation, and ranking/objective details. Thus, the comparison matches important resources and information but is not a one-variable algorithmic ablation. Equal candidate counts also do not establish equal runtime; no computational superiority claim is made here.

C. Metrics and Statistical Analysis

A collision is an event anywhere in the recorded episode. Route completion requires reaching the goal and remaining collision-free for the episode. Corridor completion additionally requires $|e_t| \leq 1.75$ m at every recorded step. Episode tracking error is

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} e_t^2}. \tag{7}$$

This is conventional episode RMSE, not the failure-penalized metric in the supplied LANS path-tracking manuscript. Since collisions can shorten an episode, RMSE is interpreted together with collision and completion outcomes.

Binary rate uncertainty uses per-condition Wilson 95% intervals. Supported continuous-mean and paired-difference intervals use 5,000 traffic-case bootstrap draws. Binary comparisons use exact paired-discordance tests; RMSE uses two-sided Wilcoxon signed-rank tests with the usual symmetry assumption for a location interpretation. We retain Holm correction [5] over all 3,440 contrasts/outcomes in the parent main-confirmation analysis, including comparisons outside the conference subset. Correction is not recomputed on a favorable selected subset. Actor-seed results remain separate for inference. Figure lines and endpoint tables average actor-condition estimates only for display; ranges across three actors are not confidence intervals.

The empirical delay threshold is the largest tested delay for which collision rate is at most 1% and corridor completion is at least 90% at that delay and every lower grid point. It is a finite-grid endpoint, not a continuous-delay stability bound. A passing value of 1,000 ms is right-censored by the tested

grid. Even zero observed collisions in 100 cases has a Wilson 95% upper bound near 3.7%, so passing does not establish a population collision probability below 1%.

VI. Results

A. LANS-RISK versus Nominal MPC

All three LANS-RISK actors and nominal MPC attain the same observed contiguous passing thresholds: Highway 1,000 ms, Merge 700 ms, Intersection 1,000 ms, and Roundabout 500 ms. Figure 2 shows that no delay-threshold advantage is established. Identical thresholds should not be read as statistical equivalence or identical episode trajectories.

At 1,000 ms, both methods complete the corridor in all Highway cases, while their actor-mean corridor rates are 35% versus 34% on Merge, 99% versus 99% on Intersection, and zero for both on Roundabout (Table I). LANS-RISK has lower mean RMSE on Merge and Roundabout, while MPC has lower mean RMSE on Highway and Intersection. These descriptive differences do not support a universal tracking winner. In particular, a small RMSE advantage on a scenario with poor corridor completion is not evidence that the task was successfully solved.

Across the 60 per-actor/scenario/delay LANS–MPC corridor-completion contrasts, nine have a positive LANS-minus-MPC mean difference, none has a negative mean difference, and none is significant after the retained study-wide Holm correction. Absence of significance is not a non-inferiority or equivalence result. We did not specify an equivalence margin or conduct a dedicated equivalence design.

B. Contribution of Queue Prediction

Disabling internal queue prediction reduces corridor-completion rates in 45 of the 60 paired actor/scenario/delay contrasts; all 45 remain significant after Holm correction. No contrast has a higher corridor-completion rate for the no-prediction variant. These counts are correlated comparisons over shared cases and actors, not 60 independent replications.

The endpoint behavior illustrates the mechanism's limitations and value. At 1,000 ms, removing prediction yields zero corridor completion in every scenario. Mean collision rates rise to 25% on Intersection and 100% on Roundabout, while Highway and Merge can have zero collisions yet still fail corridor completion. This combination explains why collision-only reporting would be misleading. The physical queue remains active in the ablation; the degradation concerns planning from the wrong effective initial state within this implementation.

C. Contribution of the Actor and Supervision Bundle

Removing the actor leaves corridor-completion rates unchanged in all 60 per-actor/scenario/delay comparisons. Table II shows mixed tracking effects at 1,000 ms: the no-actor variant has slightly lower mean RMSE on Intersection and Roundabout, but higher mean RMSE on Highway and Merge. Therefore, the present evidence does not establish a general completion benefit from the learned prior. This finding is retained rather than interpreting the presence of PPO as evidence of its usefulness.

Against PPO-only, LANS-RISK has higher corridor-completion rates in 49 of 60 contrasts, with 45 significant after Holm correction. The improvement applies to the full supervision bundle. PPO-only has zero corridor completion at the largest tested delay in every scenario and high collision rates on some scenarios. Nevertheless, comparing a supervised trajectory planner with an unsupervised policy cannot isolate which individual supervisory element causes every observed improvement.

VII. Discussion and Limitations

The main implication is narrower than a generic learned-control superiority claim. Queue-aware planning is consequential within the tested supervisor, while the learned proposal has no observed corridor-completion advantage over its matched removal. A strong model-based baseline matches the empirical delay threshold. These results motivate treating the actor as an optional proposal mechanism whose value must be measured, rather than assumed.

The cold FIFO also imposes a control-authority limitation. Early queued commands may move the vehicle outside the reference corridor before a newly computed correction can act. Increasing search effort at the current decision cannot alter commands that have already entered the queue. The threshold is therefore partly a property of the complete protocol, initial queue, scenario, and reference geometry, not solely of the controller's optimization quality.

Several boundaries limit generalization. The benchmark uses synthetic disc vehicles, a finite horizon, structured state access, fixed actors, and a known command queue. It does not establish real-road safety, robustness to arbitrary uncertainty, or performance with perception failures. Sampling only finitely many trajectories cannot certify feasibility; retaining a plan does not create a safe terminal set. Information about queue-prefix violations is diagnostic and should not be advertised as a guarantee of recovery.

The analysis preserves failures, paired cases, and frozen source/model hashes. Nonetheless, raw logs do not contain full per-step trajectories for independently reconstructing every recorded episode RMSE; aggregate statistics are recomputed from the saved episode-level values and the frozen metric implementation is retained. Confidence intervals are marginal, actor repetitions share traffic cases, and the fixed set of three actor seeds does not characterize all possible training outcomes. Shared-session episode timings are not evidence of real-time execution guarantees.

VIII. Conclusion

We presented and evaluated a queue-aware, trajectory-retaining supervisor for frozen learned driving policies under delayed command execution. A 22,000-record confirmation subset supports the importance of queue prediction within this design, while revealing no corridor-completion benefit

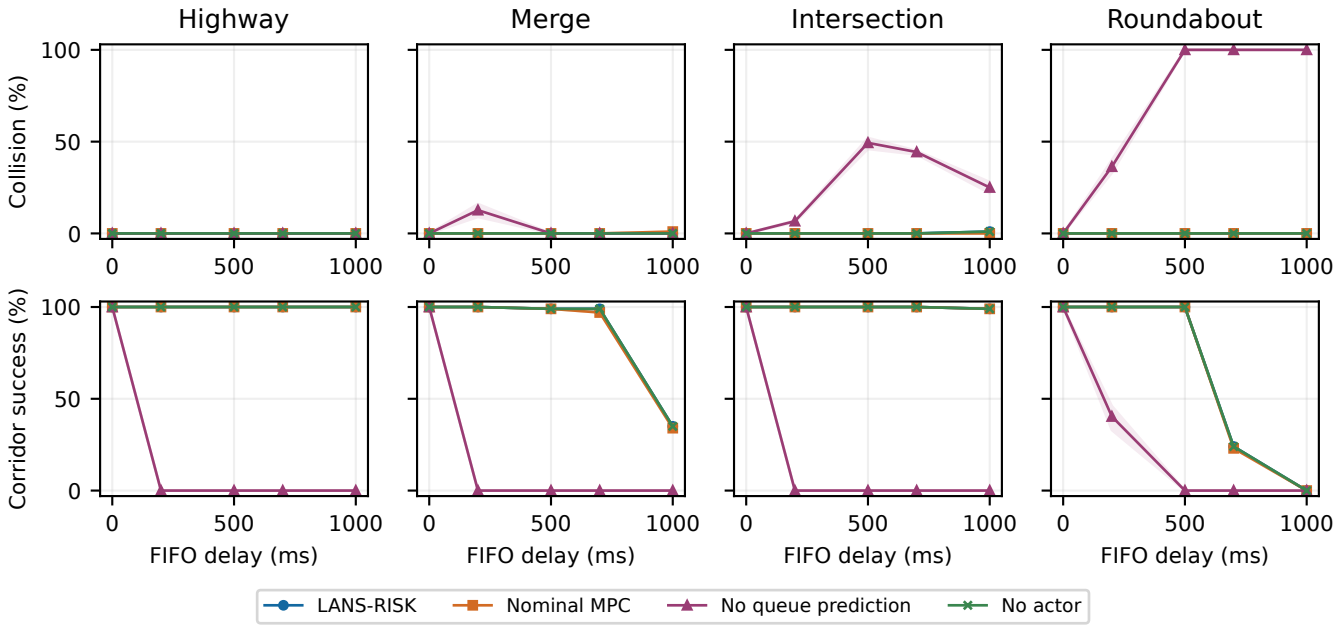


Fig. 2. All four scenarios and five main-confirmation delays. Lines show actor-seed means where applicable; shading shows the actor minimum–maximum range, not a confidence interval. Actor-free MPC and no-actor control use one run per traffic case. Overlapping lines are retained. PPO-only endpoint results are included in Table II; its complete grid is supplied in the evidence files.

TABLE I

MAIN-CONFIRMATION OUTCOMES AT 1,000 MS. LANS-RISK ENTRIES ARE DESCRIPTIVE AVERAGES OF THREE ACTOR-CONDITION MEANS; MPC USES THE SAME 100 TRAFFIC CASES ONCE. COLLISION (C) AND CORRIDOR COMPLETION (CC) ARE PERCENTAGES.

	LANS-RISK			Nominal MPC		
Scenario	C (%)	CC (%)	RMSE (m)	C (%)	CC (%)	RMSE (m)
Highway	0.0	100.0	0.080	0.0	100.0	0.025
Merge	0.0	35.0	0.378	1.0	34.0	0.416
Intersection	1.0	99.0	0.043	0.0	99.0	0.031
Roundabout	0.0	0.0	0.927	0.0	0.0	0.939

TABLE II

FOCUSED REMOVALS AND PPO-ONLY AT 1,000 MS. C: COLLISION; CC: CORRIDOR COMPLETION. ACTOR-SEED AVERAGING IS DESCRIPTIVE; PER-SEED ESTIMATES AND INTERVALS ARE SUPPLIED SEPARATELY.

Scenario	Variant	C (%)	CC (%)	RMSE (m)
Highway	LANS-RISK	0.0	100.0	0.080
Highway	No queue prediction	0.0	0.0	14.599
Highway	No actor	0.0	100.0	0.082
Highway	PPO only	0.0	0.0	11.060
Merge	LANS-RISK	0.0	35.0	0.378
Merge	No queue prediction	0.0	0.0	16.687
Merge	No actor	0.0	35.0	0.392
Merge	PPO only	0.0	0.0	35.224
Intersection	LANS-RISK	1.0	99.0	0.043
Intersection	No queue prediction	25.0	0.0	11.149
Intersection	No actor	1.0	99.0	0.032
Intersection	PPO only	19.0	0.0	14.573
Roundabout	LANS-RISK	0.0	0.0	0.927
Roundabout	No queue prediction	100.0	0.0	3.025
Roundabout	No actor	0.0	0.0	0.927
Roundabout	PPO only	99.3	0.0	3.687

from the actor prior and no observed delay-threshold advantage over matched nominal MPC. Tracking differences are scenario-dependent, and failures remain visible. The resulting contribution is an empirically characterized supervision design and a reproducible comparison, with explicit limits on novelty, safety, and generalization.

REFERENCES

[1] D. Kalaria, Q. Lin, and J. M. Dolan, "Delay-aware robust control for safe autonomous driving and racing," arXiv:2208.13856, 2022.

[2] K. P. Wabersich and M. N. Zeilinger, "A predictive safety filter for learning-based control of constrained nonlinear dynamical systems," *Automatica*, vol. 129, Art. no. 109597, 2021.

[3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[4] R. Ali and J. Tian, "Latency-aware path-tracking control for autonomous vehicles via analytical feedback and PPO-based residual compensation," author-supplied manuscript, undated.

[5] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.